

# MATH 301: Homework 2

Aren Vista

February 15, 2026

## Question 1: Section 2.1-10

**Part (a):** If  $a < b$  and  $c \leq d$ , prove that  $(a + c) < (b + d)$

Suppose  $a < b$  and  $c \leq d$ . Consider the following:

**Recall:** (*Theorem*) *Rules of Inequality*

Let  $a, b, c \in \mathbb{R}$ . The following statements must hold:

- $(a > b) \wedge (b > c) \implies a > c$
- $(a > b) \implies (a + c) > (a + b)$
- $(a > b) \wedge (c > 0) \implies ca > cb$
- $(a > b) \wedge (c < 0) \implies ca < cb$

Applying the theorem above, consider the following:

$$\begin{aligned} a &< b \\ a + c &< b + c \\ b + c &\leq b + d \\ a + c &< b + c \leq b + d \\ a + c &< b + d \end{aligned}$$

**Part (b):** If  $0 < a < b$  and  $0 \leq c \leq d$ , prove that  $0 \leq ac \leq bd$

Suppose  $0 < a < b$  and  $0 \leq c \leq d$

**Recall:** (*Theorem*) *Rules of Inequality*

Let  $a, b, c \in \mathbb{R}$ . The following statements must hold:

- $(a > b) \wedge (b > c) \implies a > c$
- $(a > b) \implies (a + c) > (a + b)$
- $(a > b) \wedge (c > 0) \implies ca > cb$
- $(a > b) \wedge (c < 0) \implies ca < cb$

Ordering If  $a - b \in P$ , we write  $a > b$  (or  $b < a$ ).

If  $a - b \in P \cup \{0\}$ , we write  $a \geq b$  (or  $b \leq a$ ).

For any two real numbers  $a, b$  exactly one of the following holds:

$$a > b, a < b, a = b$$

**Recall:** *Properties of Positive Sets*

Let  $P \neq \emptyset, P \subseteq \mathbb{R}$  be called the set of positive real numbers.

1. If  $a, b \in P$  then  $a + b \in P$ .
2. If  $a, b \in P$  then  $a \cdot b \in P$ .
3. *Trichotomy Property:* If  $a \in \mathbb{R}$ , then exactly one of the following must be true:

$$a \in P, \quad a = 0, \quad -a \in P$$

Applying the theorems above, observe the following:

$$(0 < a) \wedge (0 \leq c) \implies (0 \leq ac)$$

Thus the following statements hold:

$$0 \leq c \leq d$$

$$0 \leq bc \leq bd$$

$$0 < a < b$$

$$0 \leq ac \leq bc \leq db$$

$$0 \leq ac \leq db$$

## Question 2: Section 2.1-11

**Part (a):** If  $(a > 0)$ , then  $(1/a > 0)$  and  $\frac{1}{1/a} = a$

Suppose  $a > 0$  and  $\frac{1}{a} \not> 0$ . Consider the following:

$$\frac{1}{a} \not> 0$$

$$\left(\frac{1}{a}\right) \cdot a \not> 0 \cdot a$$

$$\left(\frac{1}{a}\right) \cdot a \not> 0$$

$$1 \not> 0$$

Clearly  $1 > 0$  thus this is a contradiction.

Thus we must accept the original proposition:  $a > 0 \implies \frac{1}{a} > 0$ .

Prove  $\frac{1}{1/a} = a$ :

$$\frac{1}{a} \cdot a = 1 \quad \text{Multiplicative Inverse}$$

$$\frac{1}{a} \cdot \frac{1}{\frac{1}{a}} = 1 \quad \text{Multiplicative Inverse}$$

$$a \cdot \frac{1}{a} \cdot \frac{1}{\frac{1}{a}} = 1 \cdot a \quad \text{Multiply } a \text{ on both sides}$$

$$1 \cdot \frac{1}{\frac{1}{a}} = 1 \cdot a \quad \text{Multiplicative Inverse}$$

$$\frac{1}{\frac{1}{a}} = a \quad \text{Multiplicative Identity}$$

Thus we can say  $\frac{1}{1/a} = a$

Now we have proven if  $(a > 0)$ , then  $(1/a > 0)$  and  $\frac{1}{1/a} = a$ .

**Part (b):** If  $(a < b)$ , then  $a < \frac{1}{2}(a + b) < b$

Consider the following:

$$a = a$$

$$\frac{1}{2}a = \frac{1}{2}a$$

$$\frac{1}{2}a + \frac{1}{2}a = \frac{1}{2}a + \frac{1}{2}a$$

$$\frac{1}{2}(a + a) = \frac{1}{2}(a + a)$$

$$\frac{1}{2}(2a) = \frac{1}{2}(a + a)$$

$$\frac{1}{2}2(a) = \frac{1}{2}(a + a)$$

$$a = \frac{1}{2}(a + a)$$

$$a < b \implies a + a < a + b$$

$$a < b \implies a + b < b + b$$

$$a < \frac{1}{2}(a + b) < b$$

### Question 3: Sec 2.1-13

If  $a, b \in \mathbb{R}$ , show that  $a^2 + b^2 = 0$  iff  $a = 0$  and  $b = 0$ .

**Case 1:**  $\rightarrow$

**Prove:** If  $a, b \in \mathbb{R}$ , show that  $a^2 + b^2 = 0$  implies  $a = 0$  and  $b = 0$ .

Consider the contrapositive statement,  $a \neq 0$  or  $b \neq 0$  implies  $a^2 + b^2 \neq 0$ :

Observe the following:

If  $a \in \mathbb{P}$

$$a \in \mathbb{P} \implies a^2 = a \cdot a \in \mathbb{P} \implies a^2 > 0$$

If  $-a \in \mathbb{P}$ : (Consider  $\forall n \in \mathbb{R}, (n - n) \cdot (-n) = 0$ )

$$\begin{aligned}
n \cdot (-n) + (-n)(-n) &= 0 \\
(-n) + (-n)(-n) &= 0 \\
\implies (-n)(-n) &= n^2 \in \mathbb{P}
\end{aligned}$$

Therefore,  $-a^2 \in \mathbb{P}$ . The same logic can be applied to  $b$ .

$$\text{Observe If } a^2, b^2 \in \mathbb{P} \implies a^2 + b^2 \in \mathbb{P}.$$

Therefore we have proven the contrapositive statement if  $a \neq 0$  or  $b \neq 0$  implies  $a^2 + b^2 \neq 0$

**Case 2:**  $\leftarrow$

**Prove:** If  $a, b \in \mathbb{R}$ , show that  $a = 0$  and  $b = 0$  implies  $a^2 + b^2 = 0$ .  
Consider the following:

$$0 \cdot 0 = 0 \implies \text{If } x = 0, \text{ then } x^2 = x \cdot x = 0 \cdot 0 = 0$$

As  $a = b = 0$  then naturally if  $a = 0$  and  $b = 0$  implies  $a^2 + b^2 = 0$

### Question 4: Section 2.1-14

If  $0 \leq a < b$ , show that  $a^2 \leq ab < b^2$ . Show by example that it does **not** follow that  $a^2 < ab < b^2$ .  
Let  $a = 0, b = 1$ . Thus  $0 \leq 0 < 1$  satisfy  $a^2 \leq ab < b^2$ . Observe  $a^2 < ab < b^2$  does not hold as  $0 < 0 < 1$  is false as  $0 \not< 0$

### Question 5: Section 2.1-20

**Part a:** If  $0 < c < 1$ , show that  $0 < c^2 < c < 1$ .

We can define  $c < 1$  as  $c = 1 - \epsilon$ , where  $\epsilon = \frac{1}{n}$  for some  $n \in \mathbb{N}$  such that  $0 < \epsilon < 1$ .

Observe the following:

$$\begin{aligned}
1 &> 1 - \epsilon \\
c &= 1 - \frac{1}{n} \\
c^2 &= \left(1 - \frac{1}{n}\right)^2 = 1 - \frac{2}{n} + \frac{1}{n^2}
\end{aligned}$$

We want to compare  $c$  and  $c^2$ . Since  $n \geq 1$  and  $\epsilon > 0$ , multiplying the inequality  $c < 1$  by  $c$  (where  $c > 0$ ) preserves the sign:

$$\begin{aligned}
c \cdot c &< 1 \cdot c \\
c^2 &< c
\end{aligned}$$

Thus, combining with our initial range:

$$0 < c^2 < c < 1$$

**Part b: If  $1 < c$ , show that  $1 < c < c^2$ .**

We can define  $c > 1$  as  $c = 1 + \epsilon$ , where  $\epsilon = \frac{1}{n}$  for some  $n \in \mathbb{N}$ .

Observe the following:

$$\begin{aligned}1 &< 1 + \epsilon \\c &= 1 + \epsilon \\c^2 &= (1 + \epsilon)^2 \\&= 1 + 2\epsilon + \epsilon^2\end{aligned}$$

Since  $\epsilon > 0$ , it must be that  $\epsilon < 2\epsilon + \epsilon^2$ . Therefore:

$$\begin{aligned}1 &< 1 + \epsilon < 1 + 2\epsilon + \epsilon^2 \\1 &< c < c^2\end{aligned}$$

Since we defined  $\epsilon > 0$ , the statement  $1 < c < c^2$  holds.

## Question 6: Section 2.1-23

If  $a > 0$ ,  $b > 0$ , and  $n \in \mathbb{N}$ , show that  $a < b$  iff  $a^n < b^n$  (induct.)

$\forall a, b \in \mathbb{R}, a > 0, b > 0$  and  $n \in \mathbb{N}$ .

Consider  $P(n) : a^n < b^n \iff a < b$

Observe  $P(1)$ :

$$\begin{aligned}a^1 &< b^1 \implies a < b \\a &< b \implies a^1 < b^1\end{aligned}$$

Suppose  $P(k), k \in \mathbb{N}$ :

$$a^k < b^k \implies a < b$$

Observe the following:

$$\begin{aligned}a^k &< b^k \\a \cdot a^k &< b \cdot a^k & \text{Eq 1} \\b \cdot a^k &< b \cdot b^k & \text{Eq 2}\end{aligned}$$

From Eq (1,2), we can derive the following

$$\begin{array}{r}a^{k+1} < b \cdot a^k \\+ \quad b \cdot a^k < b^{k+1} \\ \hline a^{k+1} + (ba^k) < (ba^k) + b^{k+1}\end{array}$$

Thus we can apply algebra to derive the final statement.

$$\begin{aligned}a^{k+1} + (ba^k) + (-ba^k) &< (ba^k) + (-ba^k) + b^{k+1} \\a^{k+1} &< b^{k+1}\end{aligned}$$

Thus we conclude  $a^k < b^k \implies a^{k+1} < b^{k+1}$ . Therefore  $a < b$  iff  $a^n < b^n$ .

## Question 7: Section 2.1-25

Assuming the existence of roots, show that  $c > 1 \implies c^{1/m} < c^{1/n}$  iff  $m > n$

### Lemma 1: Integer Powers of $c > 1$

First, we establish that for base  $c > 1$ , larger integer exponents yield larger values. We show  $c^b > c^a \iff b > a$  for  $a, b \in \mathbb{N}$ .

**Forward** ( $b > a \implies c^b > c^a$ ): Let  $b > a$ . Then  $b - a = k \in \mathbb{N}$  where  $k \geq 1$ . Using Bernoulli's Inequality for  $c = 1 + x$  where  $x > 0$ :

$$c^k = (1 + x)^k \geq 1 + kx > 1$$

Multiplying both sides by  $c^a$  (which is positive):

$$c^k \cdot c^a > 1 \cdot c^a \implies c^{a+k} > c^a \implies c^b > c^a$$

**Reverse** ( $c^b > c^a \implies b > a$ ): We proceed by contradiction. Assume  $c^b > c^a$  but  $b \leq a$ .

$$\text{If } b = a \implies c^b = c^a \quad (\text{Contradiction})$$

$$\text{If } b < a \implies a - b = k \in \mathbb{N}, k \geq 1$$

$$\implies c^a > c^b \quad (\text{By Forward proof})$$

$$(\text{Contradiction}) \therefore c^b > c^a \iff b > a$$

### Lemma 2: Roots of $c > 1$ exceed 1

Let  $c > 1$ . Let  $y = c^{1/k}$  for some  $k \in \mathbb{N}$ . We claim  $y > 1$ .

*Proof.* Assume for contradiction that  $y \leq 1$ . Since  $y > 0$ , raising both sides to the power of  $k$ :

$$y \leq 1 \implies y^k \leq 1^k \implies c \leq 1$$

This contradicts the premise  $c > 1$ . Therefore,  $c^{1/k} > 1$ . □

### Main Proof

Let  $N = mn$ . Let  $b = c^{1/N} = c^{1/mn}$ . From **Lemma 2**, we know  $b > 1$ . We can rewrite the terms  $c^{1/m}$  and  $c^{1/n}$  using base  $b$ :

$$c^{1/m} = (c^{\frac{1}{mn}})^n = b^n$$

$$c^{1/n} = (c^{\frac{1}{mn}})^m = b^m$$

We wish to show  $c^{1/m} < c^{1/n} \iff m > n$ . Substituting the expressions in terms of  $b$ :

$$b^n < b^m \iff m > n$$

Since  $b > 1$ , we apply **Lemma 1**:

$$b^m > b^n \iff m > n$$

This is exactly the statement we needed to show.

### Question 9: Section 2.2-1

**Part (a):**  $a, b \in \mathbb{R} \implies |a| = \sqrt{a^2}$

Recall:  $|a|^2 = |a| \cdot |a| = |aa| = |a^2| = a^2 \implies$

$$|a|^2 = a^2$$

$$\sqrt{|a|^2} = \sqrt{a^2}$$

$$|a| = \sqrt{a^2}$$

**Part (b):**  $a, b \in \mathbb{R} \implies |a/b| = |a|/|b|$

WLOG: Let  $a = 1$

$$\begin{cases} |1/b| = 1/b = 1/|b| \implies 1/b > 0 & b > 0 \\ |1/b| = -(1/b) = 1/(-b) = 1/|b| \implies 1/b < 0 & b < 0 \end{cases}$$

$$\therefore |a/b| = |a|/|b|$$

### Question 9: Section 2.2-10

Prove:  $a, b \in \mathbb{R} \implies |a + b| = |a| + |b| \iff ab \geq 0$

**Forward:**  $|a + b| = |a| + |b| \implies ab \geq 0$

For the sake of contradiction assume:  $|a + b| = |a| + |b| \implies ab < 0$

Let  $a = 1, b = -1$

$$|1 + (-1)| = |0| = 0$$

$$|1| + |(-1)| = 1 + 1 = 2$$

$$0 \neq 2$$

(Contradiction)  $\implies (|a + b| = |a| + |b| \implies ab \geq 0)$

**Reverse:**  $ab \geq 0 \implies |a + b| = |a| + |b|$

### Question 9: Section 2.2-12

### Question 9: Section 2.2-17